

Statement Card for the Akagera Group for the Tri Ai Project

This project will develop a Nigerian agriculture and climate focused small language model that transforms locally relevant agricultural and climate knowledge into accessible, timely, and actionable advice for rural farmers. By improving access to context-specific information on weather, climate risks, crops, pests, soil, and sustainable farming practices, the project aims to strengthen farm-level decision-making, accelerate adoption of climate-smart agriculture, reduce avoidable production losses, and enhance the resilience and livelihoods of smallholder farming communities. In the long term, the project will contribute to food security, climate adaptation, digital inclusion, and the development of locally relevant AI technologies for Nigeria.

Impact Dimension	Expected Impact
Project	Small Language Model for Agriculture and Climate Advisory Services for Rural Farmers in Nigeria
Core Problem	Rural farmers often have limited access to timely, localized, understandable, and actionable agricultural and climate information.
Target Beneficiaries	Smallholder farmers, extension workers, agricultural cooperatives, rural communities, NGOs, policymakers, and agricultural educators.
Primary Intervention	Development of a domain-specific small language model trained on Nigerian agriculture, climate, indigenous/local knowledge, farming practices, and context-specific advisory data.
Immediate Output	A reliable, resource-efficient AI system capable of providing agriculture and climate-related information in accessible language and potentially selected Nigerian/local languages.
Short-Term Impact	Improved access to agricultural knowledge, climate information, pest and disease guidance, weather-informed decision-making, and sustainable farming recommendations.
Medium-Term Impact	Improved farm-level decision-making, adoption of climate-smart agricultural practices, more efficient use of farm inputs, and stronger farmer–extension service interaction.
Long-Term Impact	Increased resilience of rural farming communities to climate variability, improved agricultural productivity and livelihoods, and strengthened food-security outcomes in Nigeria.
Climate Impact	Supports adaptation to drought, flooding, changing rainfall patterns, heat stress, soil degradation, and other climate-related agricultural risks.

Economic Impact	Potential reduction in avoidable production losses and input wastage while improving productivity, income opportunities, and the economic resilience of smallholder farmers.
Social Impact	Democratizes access to agricultural expertise by making relevant information available to underserved and digitally marginalized farming communities.
Knowledge Impact	Creates a Nigerian-focused agricultural and climate knowledge resource and contributes locally relevant training data for future AI research and development.
Technology Impact	Demonstrates how smaller, domain-specific AI models can provide useful services in resource-constrained environments where large general-purpose models may be costly or difficult to deploy.
Equity & Inclusion	Prioritizes rural and underserved populations and can be designed for low-bandwidth environments, voice interaction, and local-language accessibility.
Key Success Indicators	Farmer adoption; advisory accuracy; reduction in misinformation; improvement in climate-risk awareness; adoption of climate-smart practices; user satisfaction; productivity indicators; and measurable changes in farm decision-making.
Ultimate Impact	A locally grounded AI-enabled agricultural information ecosystem that strengthens the capacity of Nigerian smallholder farmers to make informed, climate-resilient and sustainable farming decisions.

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